

Tutorial – 1

1 : Write a program to print “Hello World”.

CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Tutroial
{
    internal class tirth
    {
        public static void Main(string[] args)
        {
            Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI");
            Console.WriteLine("Hello World"); //console is using for print
        }
    }
}
```

OUTPUT:

```
C:\WINDOWS\system32\cmd. x + v
*****
** Visual Studio 2026 Developer Command Prompt v18.7.2
** Copyright (c) 2026 Microsoft Corporation
*****

C:\Users\A\source\repos\Tutroial>csc tirth.cs
Microsoft (R) Visual C# Compiler version 5.7.0-1.26254.3 (c86410db39ebb45ae344fb1dec588b1c7802f629)
Copyright (C) Microsoft Corporation. All rights reserved.

C:\Users\A\source\repos\Tutroial>tirth.exe
24SOECE11081-SANYAJA TIRTH KALPESHBHAI
Hello World

C:\Users\A\source\repos\Tutroial>|
```

2 : Design your profile page as given below.

\$\$

Name: your name

DOB: 15/10/1991

Address: 4, xyx society,
Kalawad Road

City: Rajkot

Pincode: 360 001

State: Gujarat

Country: India

Email: abc@ymail.com

\$\$

CODE:

using System;

namespace Tutorial

{

internal class Program

{

static void Main(string[] args)

{

Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI");

Console.WriteLine("Name: Sanyaja Tirth Kalpeshbhai");

Console.WriteLine("DOB: 22/07/2007");

Console.WriteLine("Address: Vrundavan Society,");

Console.WriteLine("Sardhar");

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```
Console.WriteLine("City:Sardhar");
Console.WriteLine("Pincode: 360025");
Console.WriteLine("State: Gujarat");
Console.WriteLine("Country: India");
Console.WriteLine("Email: tirth@gmail.com");
    }
}
```

OUTPUT:

3 : Find out whether the given number is odd or even.

CODE:

using System;

namespace

Tutorial

```
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C:\Users\A\source\repos\Tutroial>csc tirth.cs
Microsoft (R) Visual C# Compiler version 5.7.0-1.26254.3 (c86410db39ebb45ae344fb1dec588b1c7802f629)
Copyright (C) Microsoft Corporation. All rights reserved.

C:\Users\A\source\repos\Tutroial>tirth.exe
24SOECE11081-SANYAJA TIRTH KALPESHBHAI
Name: Sanyaja Tirth Kalpeshbhai
DOB: 22/07/2007
Address: Vrundavan Society,
Sardhar
City:Sardhar
Pincode: 360025
State: Gujarat
Country: India
Email: tirth@gmail.com

C:\Users\A\source\repos\Tutroial>
```

```
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI");
        }
    }
}
```

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```
Console.WriteLine("Enter number: ");
int num = Convert.ToInt32(Console.ReadLine());

if (num % 2 == 0) // Even check
    Console.WriteLine("Number is Even");
else // Odd case
    Console.WriteLine("Number is Odd");
}
```

OUTPUT:

```
*****
** Visual Studio 2026 Developer Command Prompt v18.7.2
** Copyright (c) 2026 Microsoft Corporation
*****

C:\Users\A\source\repos\Tutroial>csc tirth.cs
Microsoft (R) Visual C# Compiler version 5.7.0-1.26254.3 (c86410db39ebb45ae344fb1dec588b1c7802f629)
Copyright (C) Microsoft Corporation. All rights reserved.

C:\Users\A\source\repos\Tutroial>tirth.exe
24SOECE11081-SANYAJA TIRTH KALPESHBHAI
Enter number: 5
Number is Odd

C:\Users\A\source\repos\Tutroial>
```

4 : Rearrange the given code to correct the program. The resultant program will be to input a number and print whether the given number is odd or even.

```
namespace ConsoleApplication1
{
    {
        static void Main(string[] args)
        {
            int x;
            Console.WriteLine("Enter Number : ");
            x = Convert.ToInt32(str);
            Console.WriteLine("Number is Even");
            else
            Console.Read();
                string str = Console.ReadLine();
            if (x % 2 == 0)
```

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```
        Console.WriteLine("Number is Odd");
    }
}
}
class Program
using System;
```

Output:

Enter Number: 10
Number is Even

CODE:

```
using System;

namespace ConsoleApplication1
{
    class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI");

            Console.Write("Enter Number : ");
            string str = Console.ReadLine();
            int x = Convert.ToInt32(str);    // convert to integer

            if (x % 2 == 0)
                Console.WriteLine("Number is Even");
            else
                Console.WriteLine("Number is Odd");

        }
    }
}
```

OUTPUT:

```
C:\WINDOWS\system32\cmd. x + v
*****
** Visual Studio 2026 Developer Command Prompt v18.7.2
** Copyright (c) 2026 Microsoft Corporation
*****

C:\Users\A\source\repos\Tutroial>
C:\Users\A\source\repos\Tutroial>csc pattern.cs
Microsoft (R) Visual C# Compiler version 5.7.0-1.26254.3 (c86410db39ebb45ae344fb1dec588b1c7802f629)
Copyright (C) Microsoft Corporation. All rights reserved.

C:\Users\A\source\repos\Tutroial>pattern.exe
24SOECE11081-SANYAJA TIRTH KALPESHBHAI
Enter Number : 10
Number is Even

C:\Users\A\source\repos\Tutroial>
```

5 : Write output of the program. Also write comment for each line for the following code.

```
using System;
namespace ConsoleApplication1
{
    class Program
    {
        static void Main(string[] args)
        {
            int n,fact=1;
            Console.WriteLine("Enter Number : ");
            string str = Console.ReadLine();
            n = Convert.ToInt32(str);
            for (int i = 1; i <= n; i++)
            {
                fact = fact * i;
            }
            Console.WriteLine("Factorial : {0}",fact);

            Console.Read();
        }
    }
}
```

CODE:

```
using System;

namespace ConsoleApplication1
```

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```
{
    class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI");

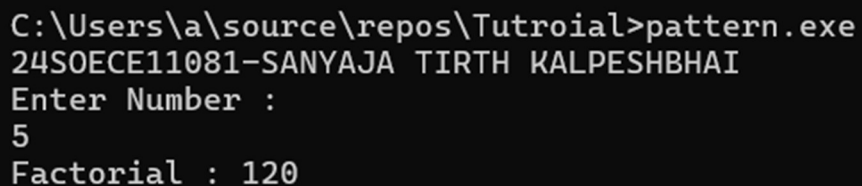
            int n, fact = 1;

            Console.WriteLine("Enter Number : ");
            string str = Console.ReadLine();
            n = Convert.ToInt32(str);

            // Loop runs from 1 to n and multiplies fact each time → factorial logic
            for (int i = 1; i <= n; i++)
            {
                fact = fact * i; // fact updated with multiplication at each step
            }

            Console.WriteLine("Factorial : {0}", fact);
        }
    }
}
```

OUTPUT:



```
C:\Users\A\source\repos\Tutorial>pattern.exe
24SOECE11081-SANYAJA TIRTH KALPESHBHAI
Enter Number :
5
Factorial : 120
```

6 : Write missing statement to get the desired output.

```
using System;
namespace ConsoleApplication1
{
    class Program
    {
        static void Main(string[] args)
```

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```
{
    int a,b,c,result;
    Console.Write("Enter Number 1: ");
//Missing statement
    a = Convert.ToInt32(str);

    Console.Write("Enter Number 2 : ");
//Missing statement
    b = Convert.ToInt32(str);

    Console.Write("Enter Number 3 : ");
    str = Console.ReadLine();
//Missing statement

    result = Sum(a, b, c);
//Missing statement

    Console.Read();
}
static int Sum(int x, int y, int z)
{
    int res;
    res = x+y+z;
    return res;
}
}
```

Output:

```
Enter Number 1 : 10
Enter Number 2 : 20
Enter Number 3 : 30
Sum : 60
```

CODE:

```
using System;

namespace ConsoleApplication1
{
    class Program
    {
        static void Main(string[] args)
        {
```

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```
Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI");
```

```
int a, b, c, result;
```

```
Console.Write("Enter Number 1 : ");
```

```
string str = Console.ReadLine(); // Missing statement → read input
```

```
a = Convert.ToInt32(str);
```

```
Console.Write("Enter Number 2 : ");
```

```
str = Console.ReadLine(); // Missing statement → read input
```

```
b = Convert.ToInt32(str);
```

```
Console.Write("Enter Number 3 : ");
```

```
str = Console.ReadLine(); // Already present
```

```
c = Convert.ToInt32(str); // Missing statement → convert to int
```

```
result = Sum(a, b, c);
```

```
Console.WriteLine("Sum : {0}", result); // Missing statement → print result
```

```
}
```

```
static int Sum(int x, int y, int z)
```

```
{
```

```
    int res;
```

```
    res = x + y + z;
```

```
    return res;
```

```
}
```

```
}
```

```
}
```

OUTPUT:

```
C:\Users\a\source\repos\Tutroial>pattern.exe
24SOECE11081-SANYAJA TIRTH KALPESHBHAI
Enter Number 1 : 10
Enter Number 2 : 20
Enter Number 3 : 30
Sum : 60

C:\Users\a\source\repos\Tutroial>|
```

7 : Predict and write the output of the given code.

```
using System;
namespace While_Loop
{
    class Program
    {
        static void Main(string[] args)
        {
            int num1,res, i;

            Console.WriteLine("Enter a number");
            num1 = Convert.ToInt32(Console.ReadLine());

            i = 1; //Initialization

            //Check whether condition matches or not
            while (i <= 10)
            {
                res = num1 * i;
                Console.WriteLine("{0} x {1} = {2}", num1, i, res);
                i++; //Increment by one
            }
            Console.ReadLine();
        }
    }
}
```

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CODE:

```
using System;
namespace While_Loop
{
    class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI");

            int num1, res, i;

            Console.WriteLine("Enter a number:");
            num1 = Convert.ToInt32(Console.ReadLine());

            i = 1; // Initialization

            // Loop runs until i <= 10
            while (i <= 10)
            {
                res = num1 * i; // multiplication logic
                Console.WriteLine("{0} x {1} = {2}", num1, i, res);

                i++; // increment
            }
            Console.ReadLine();
        }
    }
}
```

OUTPUT:

```
C:\Users\A\source\repos\Tutorial>pattern.exe
24SOECE11081-SANYAJA TIRTH KALPESHBHAI
Enter a number:
5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

8 Write a program to convert given name in upper characters.

INPUT : John F Kennedy

OUTPUT: JOHN F KENNEDY

CODE:

```
using System;
```

```
namespace ConsoleApplication1
```

```
{
```

```
    class Program
```

```
    {
```

```
        static void Main(string[] args)
```

```
        {
```

```
            Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI");
```

```
            Console.Write("Enter Name: ");
```

```
            string name = Console.ReadLine();
```

```
            // ToUpper() method converts string into uppercase characters
```

```
            string upperName = name.ToUpper();
```

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```
Console.WriteLine("Converted Name: " + upperName);
```

```
    }  
  }  
}
```

OUTPUT:

```
C:\Users\A\source\repos\Tutroial>pattern.exe  
24SOECE11081-SANYAJA TIRTH KALPESHBHAI  
Enter Name: john F KENNEDY  
Converted Name: JOHN F KENNEDY
```

9 Write a Program to convert given name in toggle case.

INPUT : JoHn F kEnNedy

OUTPUT: jOhN f KeNneDY

CODE:

```
using System;
```

```
namespace ConsoleApplication1
```

```
{
```

```
    class Program
```

```
    {
```

```
        static void Main(string[] args)
```

```
        {
```

```
            Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI");
```

```
            Console.Write("Enter Name: ");
```

```
            string name = Console.ReadLine();
```

```
            string toggleName = "";
```

```
            // Loop through each character and toggle case
```

```
            foreach (char ch in name)
```

```
            {
```

```
                if (char.IsUpper(ch))
```

```
                    toggleName += char.ToLower(ch); // convert uppercase → lowercase
```

```
                else
```

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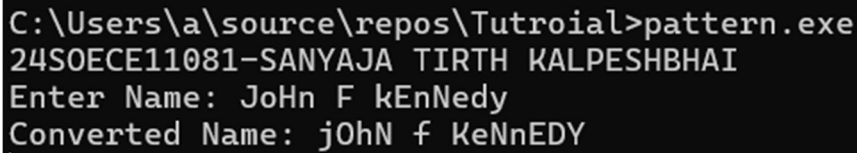
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```
toggleName += char.ToUpper(ch); // convert lowercase → uppercase  
}
```

```
Console.WriteLine("Converted Name: " + toggleName);
```

```
    }  
}  
}
```

OUTPUT:



```
C:\Users\A\source\repos\Tutorial>pattern.exe  
24SOECE11081-SANYAJA TIRTH KALPESHBHAI  
Enter Name: JoHn F kEnNedy  
Converted Name: jOhN f KeNnEDY
```

10 Write a Program which accepts mobile no as a string from the user and converts the last 5 digits into X.

INPUT : 1234567890

OUTPUT: 12345XXXXX

CODE:

```
using System;
```

```
namespace ConsoleApplication1
```

```
{
```

```
    class Program
```

```
    {
```

```
        static void Main(string[] args)
```

```
        {
```

```
            Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI");
```

```
            Console.Write("Enter Mobile Number: ");
```

```
            string mobile = Console.ReadLine();
```

```
            // Substring(0, mobile.Length - 5) → first part of number
```

```
            string masked = mobile.Substring(0, mobile.Length - 5) + "XXXXX";
```

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```
Console.WriteLine("Converted Mobile No: " + masked);
```

```
    }  
  }  
}
```

OUTPUT:

```
C:\Users\A\source\repos\Tutroial>pattern.exe  
24SOECE11081-SANYAJA TIRTH KALPESHBHAI  
Enter Mobile Number: 1234567890  
Converted Mobile No: 12345XXXXX
```

11 Write a Program which accepts name and gender from the user. Here, gender may have only 1 character, M or F.

Based on the gender prefix the name Mr. & Ms.

NAME : Hillary Clinton

GENDER : F

CODE:

```
using System;
```

```
namespace ConsoleApplication1
```

```
{
```

```
    class Program
```

```
    {
```

```
        static void Main(string[] args)
```

```
        {
```

```
            Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI");
```

```
            Console.Write("Enter Name: ");
```

```
            string name = Console.ReadLine();
```

```
            Console.Write("Enter Gender (M/F): ");
```

```
            char gender = Convert.ToChar(Console.ReadLine());
```

```
            // Check gender and prefix accordingly
```

```
            if (gender == 'M' || gender == 'm')
```

```
                Console.WriteLine("Mr. " + name);
```

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```
else if (gender == 'F' || gender == 'f')
    Console.WriteLine("Ms. " + name);
else
    Console.WriteLine("Invalid gender input");
}
}
```

OUTPUT:

```
C:\Users\A\source\repos\Tutorial>pattern.exe
24SOECE11081-SANYAJA TIRTH KALPESHBHAI
Enter Name: Hillary Clinton
Enter Gender (M/F): F
Ms. Hillary Clinton
```

12 Write a Program which accepts name from the user and prints the same

INPUT : Winston Churchill

OUTPUT: Winston Churchill

CODE:

```
using System;

namespace ConsoleApplication1
{
    class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI");

            Console.Write("Enter Name: ");
            string name = Console.ReadLine();

            Console.WriteLine("You entered: " + name);

            Console.Read();
        }
    }
}
```

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OUTPUT:

```
C:\Users\A\source\repos\Tutroial>pattern.exe
24SOECE11081-SANYAJA TIRTH KALPESHBHAI
Enter Name: Winston Churchill
You entered: Winston Churchill
```

13. Write a Program to prints the following series
0 1 1 2 3 5 8 13 21 34 55

CODE:

```
using System;

namespace ConsoleApplication1
{
    class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI");

            int n1 = 0, n2 = 1, n3;
            Console.Write(n1 + " " + n2 + " ");

            // Loop prints next 9 terms → total 11 terms
            for (int i = 2; i < 11; i++)
            {
                n3 = n1 + n2; // next term = sum of previous two
                Console.Write(n3 + " ");
                n1 = n2;
                n2 = n3;
            }
        }
    }
}
```

OUTPUT:

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```
C:\Users\A\source\repos\Tutroial>pattern.exe
24SOECE11081-SANYAJA TIRTH KALPESHBHAI
0 1 1 2 3 5 8 13 21 34 55 |
```

14 Write a Program which accepts no from the user and print the same in words.

INPUT : 98732

OUTPUT: Nine Eight Seven Three Two

CODE:

```
using System;
```

```
namespace ConsoleApplication1
```

```
{
```

```
    class Program
```

```
    {
```

```
        static void Main(string[] args)
```

```
        {
```

```
            Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI");
```

```
            Console.Write("Enter Number: ");
```

```
            string num = Console.ReadLine();
```

```
            // Loop through each character of the number string
```

```
            foreach (char ch in num)
```

```
            {
```

```
                switch (ch)
```

```
                {
```

```
                    case '0': Console.Write("Zero "); break;
```

```
                    case '1': Console.Write("One "); break;
```

```
                    case '2': Console.Write("Two "); break;
```

```
                    case '3': Console.Write("Three "); break;
```

```
                    case '4': Console.Write("Four "); break;
```

```
                    case '5': Console.Write("Five "); break;
```

```
                    case '6': Console.Write("Six "); break;
```

```
                    case '7': Console.Write("Seven "); break;
```

```
                    case '8': Console.Write("Eight "); break;
```

```
                    case '9': Console.Write("Nine "); break;
```

```
                    default: Console.Write("? "); break; // for invalid input
```

```
                }
```

```
            }
```

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```
}  
}  
}
```

OUTPUT:

```
C:\Users\a\source\repos\Tutroial>pattern.exe  
24SOECE11081-SANYAJA TIRTH KALPESHBHAI  
Enter Number: 98732  
Nine Eight Seven Three Two |
```

15 Write a Program to check whether the given no is Armstrong no or not.

CODE:

```
using System;  
  
namespace ConsoleApplication1  
{  
    class Program  
    {  
        static void Main(string[] args)  
        {  
            Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI");  
  
            Console.Write("Enter Number: ");  
            int num = Convert.ToInt32(Console.ReadLine());  
  
            int temp = num;  
            int sum = 0;  
  
            // Loop through each digit and add cube of digit  
            while (temp > 0)  
            {  
                int digit = temp % 10;        // extract last digit  
                sum += digit * digit * digit; // cube and add to sum  
                temp /= 10;                    // remove last digit  
            }  
  
            if (sum == num)  
                Console.WriteLine(num + " is an Armstrong number");  
            else  
                Console.WriteLine(num + " is not an Armstrong number");  
        }  
    }  
}
```

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```
}  
}  
}
```

OUTPUT:

```
C:\Users\A\source\repos\Tutroial>pattern.exe  
24SOECE11081-SANYAJA TIRTH KALPESHBHAI  
Enter Number: 153  
153 is an Armstrong number
```

16 Write a program to display a pattern like a right angle triangle using an asterisk
The pattern like :

```
*  
**  
***  
****
```

CODE:

```
using System;
```

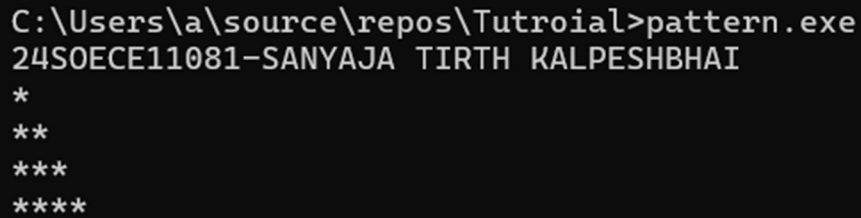
```
namespace ConsoleApplication1  
{  
    class Program  
    {  
        static void Main(string[] args)  
        {  
            Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI");  
  
            // Loop controls number of rows  
            for (int i = 1; i <= 4; i++)  
            {  
                // Inner loop prints '*' i times  
                for (int j = 1; j <= i; j++)  
                {  
                    Console.Write("*");  
                }  
                Console.WriteLine(); // move to next line  
            }  
        }  
    }  
}
```

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```
}  
}  
}
```

OUTPUT:



```
C:\Users\A\source\repos\Tutorial>pattern.exe  
24SOECE11081-SANYAJA TIRTH KALPESHBHAI  
*  
**  
***  
****
```

17. Write a Program to generate following output.

```
1  
1 2  
1 2 3  
1 2 3 4
```

CODE:

```
using System;  
  
namespace ConsoleApplication1  
{  
    class Program  
    {  
        static void Main(string[] args)  
        {  
            Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI");  
  
            // Outer loop controls number of rows  
            for (int i = 1; i <= 4; i++)  
            {  
                // Inner loop prints numbers from 1 to i  
                for (int j = 1; j <= i; j++)  
                {  
                    Console.Write(j + " ");  
                }  
                Console.WriteLine(); // move to next line  
            }  
        }  
    }  
}
```

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```
}  
}  
}
```

OUTPUT:

```
C:\Users\A\source\repos\Tutorial>pattern.exe  
24SOECE11081-SANYAJA TIRTH KALPESHBHAI  
1  
1 2  
1 2 3  
1 2 3 4
```

18 Write a program to make such a pattern like a right angle triangle with the number increased by 1.

The pattern like :

```
1  
2 3  
4 5 6  
7 8 9 10
```

CODE:

```
using System;
```

```
namespace ConsoleApplication1  
{  
    class Program  
    {  
        static void Main(string[] args)  
        {  
            Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI");  
  
            int num = 1; // starting number  
  
            // Outer loop controls number of rows  
            for (int i = 1; i <= 4; i++)  
            {  
                // Inner loop prints 'i' numbers in each row  
                for (int j = 1; j <= i; j++)  
                {  
                    Console.Write(num + " ");  

```

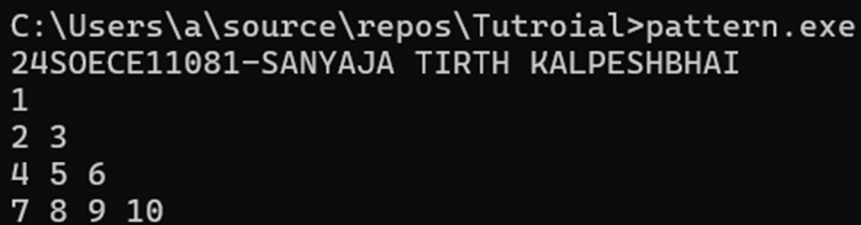
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```
        num++; // increase number by 1 each time
    }
    Console.WriteLine(); // move to next line
}

}
}
}
```

OUTPUT:



```
C:\Users\A\source\repos\Tutorial>pattern.exe
24SOECE11081-SANYAJA TIRTH KALPESHBHAI
1
2 3
4 5 6
7 8 9 10
```

19. Write a program to make such a pattern as a pyramid with an asterisk.

```
*
* *
* * *
* * * *
```

CODE:

```
using System;

namespace ConsoleApplication1
{
    class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI");

            int rows = 4;

            // Outer loop controls number of rows
            for (int i = 1; i <= rows; i++)
            {
```

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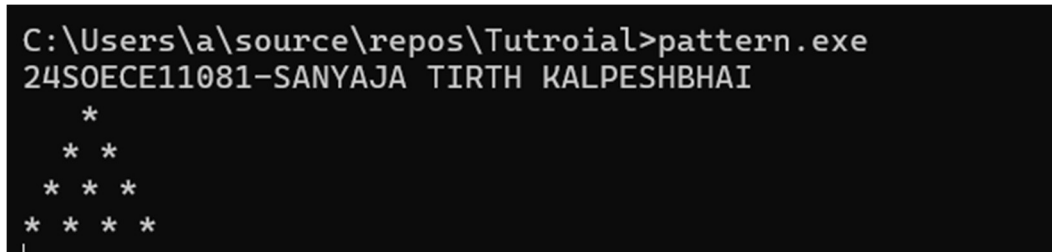
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```
// Print leading spaces → pyramid alignment
for (int space = 1; space <= rows - i; space++)
{
    Console.Write(" ");
}

// Print '*' with space between them
for (int star = 1; star <= i; star++)
{
    Console.Write("* ");
}

Console.WriteLine(); // move to next line
}
}
}
```

OUTPUT:



```
C:\Users\A\source\repos\Tutorial>pattern.exe
24SOECE11081-SANYAJA TIRTH KALPESHBHAI
 *
* *
* * *
* * * *
```

20. Write a program to make a pyramid pattern with numbers increased by 1.

```
1
2 3
4 5 6
7 8 9 10
```

CODE:

```
using System;

namespace ConsoleApplication1
{
    class Program
    {
```

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```
static void Main(string[] args)
{
    Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI");

    int num = 1; // starting number
    int rows = 4;

    // Outer loop controls number of rows
    for (int i = 1; i <= rows; i++)
    {
        // Print leading spaces for pyramid alignment
        for (int space = 1; space <= rows - i; space++)
        {
            Console.Write(" ");
        }

        // Inner loop prints numbers with space
        for (int j = 1; j <= i; j++)
        {
            Console.Write(num + " ");
            num++; // increase number by 1 each time
        }

        Console.WriteLine(); // move to next line
    }
}
```

OUTPUT:

```
C:\Users\A\source\repos\Tutroial>pattern.exe
24SOECE11081-SANYAJA TIRTH KALPESHBHAI
 1
2 3
4 5 6
7 8 9 10
```

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21. Write a program to find the sum of the series $5 + 55 + 555 + 5555 + \dots n$ terms.

Test Data :

Input the number of terms : 4

Input number : 5

Expected Output :

$5 + 55 + 555 + 5555$

The Sum is : 6170

CODE:

using System;

namespace ConsoleApplication1

{

class Program

{

static void Main(string[] args)

{

Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI");

Console.Write("Input the number of terms: ");

int n = Convert.ToInt32(Console.ReadLine());

Console.Write("Input number: ");

int num = Convert.ToInt32(Console.ReadLine());

int term = num;

int sum = 0;

Console.Write("Series: ");

for (int i = 1; i <= n; i++)

{

Console.Write(term);

if (i < n) Console.Write(" + ");

sum += term;

term = term * 10 + num; // next term logic

}

Console.WriteLine();

Console.WriteLine("The Sum is : " + sum);

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```
}  
}  
}
```

OUTPUT:

```
C:\Users\A\source\repos\Tutroial>pattern.exe  
24SOECE11081-SANYAJA TIRTH KALPESHBHAI  
Input the number of terms: 5  
Input number: 5  
Series: 5 + 55 + 555 + 5555 + 55555  
The Sum is : 61725
```

22. Write a program to display a pattern like a diamond.

```
*  
***  
*****  
*****  
*****  
*****  
*****  
*****  
***  
*
```

CODE:

```
using System;
```

```
namespace Tutroial
```

```
{  
    internal class Program1  
    {  
        static void Main(string[] args)  
        {  
            Console.WriteLine("24SOECE11081-SANYAJA TIRTH KALPESHBHAI:");  
            int row = 5;  
            //upper triangle(half)  
            for (int i = 1; i <= row; i++)  
            {  
                for (int j = i; j < row; j++)
```

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```
{
    Console.Write(" ");
}
for (int k = 1; k <= (2 * i) - 1; k++)
{
    Console.Write("*");
}
Console.WriteLine();
}
//lower triangle(half)
for (int i = row - 1; i >= 1; i--)
{
    for (int j = i; j < row; j++)
    {
        Console.Write(" ");
    }
    for (int k = 1; k <= (2 * i) - 1; k++)
    {
        Console.Write("*");
    }
    Console.WriteLine();
}
}
```

OUTPUT:

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```
C:\Users\a\source\repos\Tutroial>pattern.exe  
24SOECE11081-SANYAJA TIRTH KALPESHBHAI:
```

```
  *  
 ***  
*****  
*****  
*****  
*****  
*****  
*****  
***  
  *
```